step3_all_model_train_rf

February 6, 2026

[1]:
```
import pandas as pd
import numpy as np
import os
import joblib
import lightgbm as lgb
```

[2]:
```
from sklearn.model_selection import StratifiedGroupKFold
from sklearn.preprocessing import StandardScaler #
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from lightgbm import LGBMClassifier
```

[3]:
```
data_dir = '../../data'
input_dir = f'{data_dir}/step2_processing_outputs'
output_dir = f'{data_dir}/step3_model_outputs'

file_paths = {
    'tagt_feat': f'{input_dir}/step2_all_tagt_feat'
}
```

[4]:
```
#    DataFrame
dataframes = {}

#         pd.read_paquet
for key, file_path in file_paths.items():
    if not os.path.exists(file_path):
        print(f" :    '{file_path}'          ")
        continue

    try:
        # lineterminator='\n'
        df = pd.read_parquet(file_path)
        dataframes[key] = df
        print(f"'{key}'      DataFrame  : {df.shape}")
    except Exception as e:
        print(f"  : '{file_path}'             : {e}")
```

```
 :    '../../data/step2_processing_outputs/step2_all_tagt_feat'
```

[5]:
```
#data
df_tagt_feat = dataframes['tagt_feat']
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[5], line 2
      1 #data
----> 2 df_tagt_feat = dataframes['tagt_feat']

KeyError: 'tagt_feat'
```

[ ]:
```
#
feat_column_names = df_tagt_feat.drop(columns=['hs','Event','DO_DATE']).columns.
 ↪tolist()
```

[6]:
```
#    (X)     (y)
features = feat_column_names
target = 'Event'

X = df_tagt_feat[features]
y = df_tagt_feat[target]
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[6], line 2
      1 #    (X)     (y)
----> 2 features = feat_column_names
      3 target = 'Event'
      5 X = df_tagt_feat[features]

NameError: name 'feat_column_names' is not defined
```

[7]:
```
sgkf = StratifiedGroupKFold(n_splits=5, shuffle=True, random_state=42)

# n_splits=5     20%      test_size=0.2
# 1      train_test_split
groups = df_tagt_feat['hs']
train_idx, test_idx = next(sgkf.split(X, y, groups=groups))

#
X_train, X_test = X.iloc[train_idx].copy(), X.iloc[test_idx].copy()
y_train, y_test = y.iloc[train_idx].copy(), y.iloc[test_idx].copy()
```

```
---------------------------------------------------------------------------
```

```
NameError                                 Traceback (most recent call last)
Cell In[7], line 5
      1 sgkf = StratifiedGroupKFold(n_splits=5, shuffle=True, random_state=42)
      3 # n_splits=5     20%      test_size=0.2
      4 # 1      train_test_split
----> 5 groups = df_tagt_feat['hs']
      6 train_idx, test_idx = next(sgkf.split(X, y, groups=groups))
      8 #

NameError: name 'df_tagt_feat' is not defined
```

[8]:

```
#
fill_cols = ['lastNEUT#', 'lastWBC']

# 1.    (train)
train_medians = X_train[fill_cols].median()

# 2.
X_train[fill_cols] = X_train[fill_cols].fillna(train_medians)

# 3.
X_test[fill_cols] = X_test[fill_cols].fillna(train_medians)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[8], line 5
      2 fill_cols = ['lastNEUT#', 'lastWBC']
      4 # 1.    (train)
----> 5 train_medians = X_train[fill_cols].median()
      7 # 2.
      8 X_train[fill_cols] = X_train[fill_cols].fillna(train_medians)

NameError: name 'X_train' is not defined
```

[9]:

```
#
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

#        DataFrame
X_train_scaled_df = pd.DataFrame(X_train_scaled, columns=features,␣
 ↪index=X_train.index)
X_test_scaled_df = pd.DataFrame(X_test_scaled, columns=features, index=X_test.
 ↪index)
y_test_df = pd.DataFrame(y_test, columns=['Event'])
```

```
#evaluation
X_test_scaled_df.to_parquet(f'{output_dir}/step3_all_x_test_scaled')
X_test_scaled_df.to_excel(f'{output_dir}/step3_all_x_test_scaled.xlsx',␣
↪sheet_name='x_test_scaled')
y_test_df.to_parquet(f'{output_dir}/step3_all_y_test')
y_test_df.to_excel(f'{output_dir}/step3_all_y_test.xlsx', sheet_name='y_test')
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[9], line 3
      1 #
      2 scaler = StandardScaler()
----> 3 X_train_scaled = scaler.fit_transform(X_train)
      4 X_test_scaled = scaler.transform(X_test)
      6 #        DataFrame

NameError: name 'X_train' is not defined
```

[10]:
```
#
#
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train_scaled_df, y_train)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[10], line 4
      1 #
      2 #
      3 rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
----> 4 rf_model.fit(X_train_scaled_df, y_train)

NameError: name 'X_train_scaled_df' is not defined
```

[ ]:
```
#XGboost

#
tuned_weight = np.sqrt(ratio)

xgb_model = XGBClassifier(
    scale_pos_weight=tuned_weight,
    n_estimators=1000,
    learning_rate=0.05,
    max_depth=7,
    min_child_weight=5,
    subsample=0.8,
```

```
    colsample_bytree=0.8,
    random_state=42,
    eval_metric='aucpr',
    early_stopping_rounds=200
)

#
xgb_model.fit(
    X_train_scaled,
    y_train,
    eval_set=[(X_test_scaled, y_test)],
    verbose=False
)
```

[11]:
```
#     XGBoost
tuned_weight = np.sqrt(ratio)

# LightGBM
lgb_model = LGBMClassifier(
    scale_pos_weight=tuned_weight,
    n_estimators=1000,
    learning_rate=0.05,
    max_depth=7,
    num_leaves=31,                # LightGBM      2^max_depth
    min_child_samples=5,          # XGBoost min_child_weight
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    importance_type='gain',       #
    verbosity=-1                  #
)

#         Early Stopping
lgb_model.fit(
    X_train_scaled,
    y_train,
    eval_set=[(X_test_scaled, y_test)],
    eval_metric='auc',            # AUC
    callbacks=[
        lgb.early_stopping(stopping_rounds=200),
        lgb.log_evaluation(period=0) #       (verbose=False )
    ]
)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[11], line 2
```

```
      1 #    XGBoost
----> 2 tuned_weight = np.sqrt(ratio)
      4 # LightGBM
      5 lgb_model = LGBMClassifier(
      6     scale_pos_weight=tuned_weight,
      7     n_estimators=1000,
   (…)
     16     verbosity=-1               #
     17 )

NameError: name 'ratio' is not defined
```

[12]:
```
#
joblib.dump(rf_model,  f'{output_dir}/all_rf_model.joblib')
joblib.dump(xgb_model,  f'{output_dir}/all_xgb_model.joblib')
joblib.dump(lgb_model,  f'{output_dir}/all_lgb_model.joblib')
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
Cell In[12], line 2
      1 #
----> 2 joblib.dump(rf_model,  f'{output_dir}/all_rf_model.joblib')
      3 joblib.dump(xgb_model,  f'{output_dir}/all_xgb_model.joblib')
      4 joblib.dump(lgb_model,  f'{output_dir}/all_lgb_model.joblib')

File /opt/conda/lib/python3.11/site-packages/joblib/numpy_pickle.py:552, in
 ↪dump(value, filename, compress, protocol, cache_size)
    550         NumpyPickler(f, protocol=protocol).dump(value)
    551 elif is_filename:
--> 552     with open(filename, 'wb') as f:
    553         NumpyPickler(f, protocol=protocol).dump(value)
    554 else:

FileNotFoundError: [Errno 2] No such file or directory: '../../data/
 ↪step3_model_outputs/all_rf_model.joblib'
```